<u>OpenNebula Project</u>

- **Project Goal**
  You will build a **basic research cloud portal** where a user can request a virtual machine (VM) and use it for simple computation.
  Think of it as a **mini cloud system (very simplified AWS idea)** but implemented using OpenNebula tools.

- **Required Tasks**

1. OpenNebula Setup

- Install and configure OpenNebula
- Create at least **2 VM images** (e.g., Ubuntu + Windows or two Linux variants)

2. Simple User Access

- Create **at least 2 users** in OpenNebula
- Each user should be able to log in and launch a VM

3. VM Request

Each user should be able to:

- Choose an OS image
- Launch a VM with:
  - CPU (fixed value is OK)
  - RAM (fixed value is OK)

4. Run a Simple Task on VM

After launching the VM:

- Run a **basic task**, such as:
  - Python script
  - Simple computation
  - Data processing on a small file

5. Output Retrieval

- Show that the user can **get the result/output** from the VM
- Example:
  - Download file
  - Display result screenshot

- **Deliverables**

Each group must submit:

- Short report
- Screenshots showing:
    - VM creation
    - User access

- **Bonus Tasks**
    - Allow user to choose different CPU/RAM dynamically
    - Add **GPU option (simulation is enough)**
    - Create a simple web portal instead of using Sunstone
    - Provide **preconfigured environments** (e.g., Python + libraries)
    - Automate task execution after VM launch
- **Group Size**
    - Each group should have **2–6 students only**